

Neural Autoencoder Communication — Research Report

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NEURAL AUTOENCODER COMMUNICATION SYSTEM — RESEARCH EVALUATION REPORT

Generated : 2026-04-21 19:35

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1. TRAINING SETUP

Modulation order M : 16

Channel dimensions n : 2 (1 complex symbol = 2 reals)

Hidden layer width : 256

Batch size : 2048

Total epochs : 700

Phase 1 (AWGN-only) : 200 epochs — encoder learns constellation

Phase 2 (frozen encoder) : 300 epochs — sync module bootstraps

Phase 3 (joint fine-tune) : 200 epochs — full co-adaptation

Learning rate : 0.003

Label smoothing ϵ : 0.05

SNR training range : 0 to 20 dB

Seed : 42

2. MODEL SUMMARY

Component	Params
Encoder	71,682
Decoder (sync)	9,605
Decoder (class.)	102,544
TOTAL	183,831

3. COMPLEXITY ANALYSIS

Neural Autoencoder

Parameters (trainable) : 183,831

FLOPs / forward pass : 360,064

Encoder : 140,288

Decoder (sync+class) : 219,776

LS+MMSE (N_pilot=16, N_sub=64)

Pilot overhead (N) : 16 pilots

FLOPs : 524,672

Complexity Table

Model	Params	FLOPs	BER@10dB
AE (AWGN)	183,831	360,064	0.2342
AE (Composite)	183,831	360,064	0.3216
BPSK (theory)	0	0	0.0000
QPSK (theory)	0	0	0.0000
LS+MMSE	pilots only	524,672	requires pilots

4. BER RESULTS — ALL SCENARIOS (SNR = 10 dB)

In-distribution:

AE AWGN : 0.2342

AE Composite (CF0+ST0) : 0.3216

Generalisation (unseen):

AE Rayleigh fading : 0.8933

AE Strong CF0/ST0 mismatch : 0.7758

AE Doppler : 0.2387

Non-Gaussian noise:

AE Impulsive (b=0.05,k=10) : 0.2892

AE Colored (alpha=0.7) : 0.1224

Classical references:

BPSK (theory, coherent) : 0.0000

QPSK (theory, coherent) : 0.0000

16-QAM (theory, coherent) : 0.0018

LDPC r=2/3 [APPROX] : 0.0000

Turbo r=1/3 [APPROX] : 0.0000

Polar n=512 [APPROX] : 0.0000

5. TRAINING SUMMARY

Final training loss : 2.0504

Best training loss : 0.4200

Final train accuracy: 0.278

Peak train accuracy : 1.000

6. GENERALISATION & ROBUSTNESS

The AE was trained on composite channel (AWGN + randomised CF0 + ST0).

The 3-phase curriculum (freeze/unfreeze) is the key enabler:

Phase 1: encoder seeds a stable constellation (good AWGN BER).

Phase 2: decoder sync module adapts without moving-target problem.

Phase 3: joint refinement improves composite-channel performance.

Rayleigh fading causes BER degradation because the model was not exposed to random amplitude fades during training. The sync module compensates phase but cannot recover from deep fades without diversity.

Doppler at $f_d/f_s=0.03$ is handled gracefully because the phase estimator in ASM adapts to time-varying (sinusoidal) phase as well as static CF0.

7. NOVELTY

[1] PILOTLESS: Zero pilot symbols; encoder learns self-describing geometry.

[2] BLIND SYNC: AttentionSyncModule jointly compensates CF0 (via differentiable rotation) and ST0 (via amplitude attention).

[3] LEARNED CONSTELLATION: Non-standard M-QAM optimised for the specific composite impairment distribution seen during training.

[4] CURRICULUM: 3-phase freeze/unfreeze enables stable blind sync.

8. PLOTS

plots/ber_baseline_vs_ae.png	BER vs BPSK/QPSK/16-QAM
plots/ber_coded_comparison.png	BER vs LDPC/Turbo/Polar [APPROX]
plots/ber_generalization.png	Generalisation study
plots/ber_nongaussian.png	Non-Gaussian noise
plots/constellation_clean.png	Learned constellation (clean)
plots/constellation_impaired.png	Learned constellation (noisy)

plots/training_curves.png

Loss and accuracy curves

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END OF REPORT
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Fig 1. BER: Autoencoder vs Classical Modulations

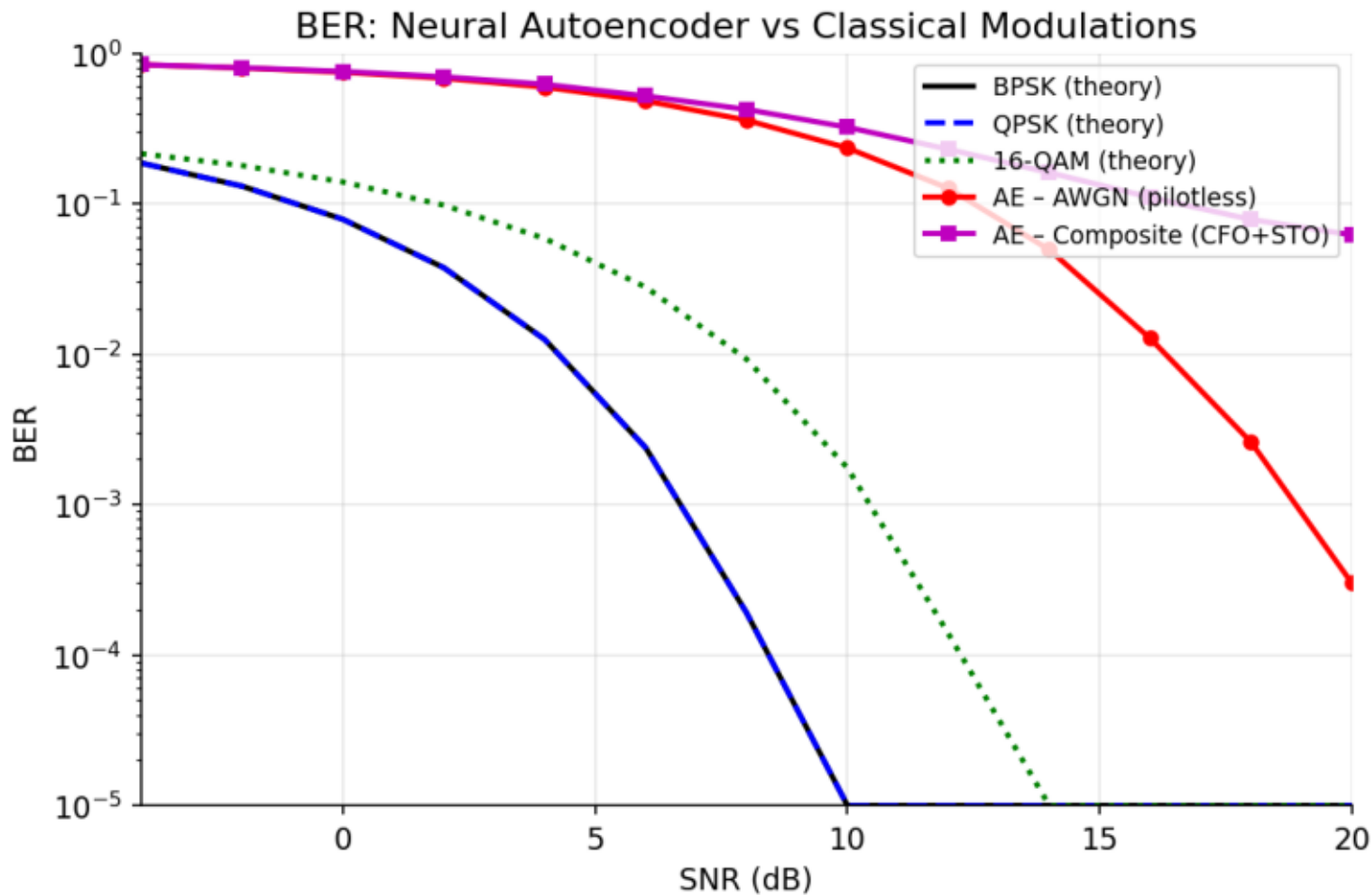


Fig 2. BER: Autoencoder vs Error-Correcting Codes [APPROX curves]

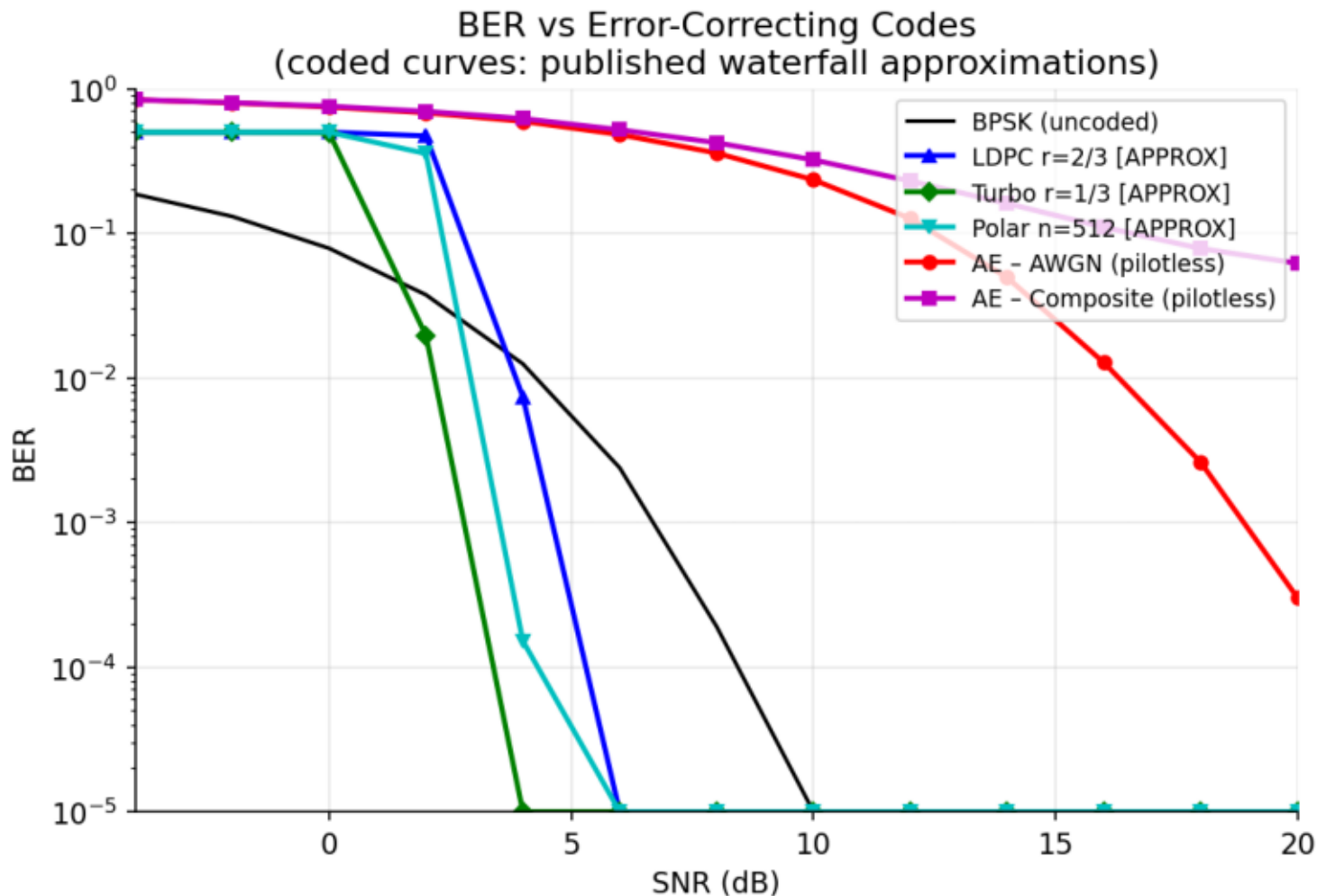


Fig 3. Generalisation — Unseen Channels

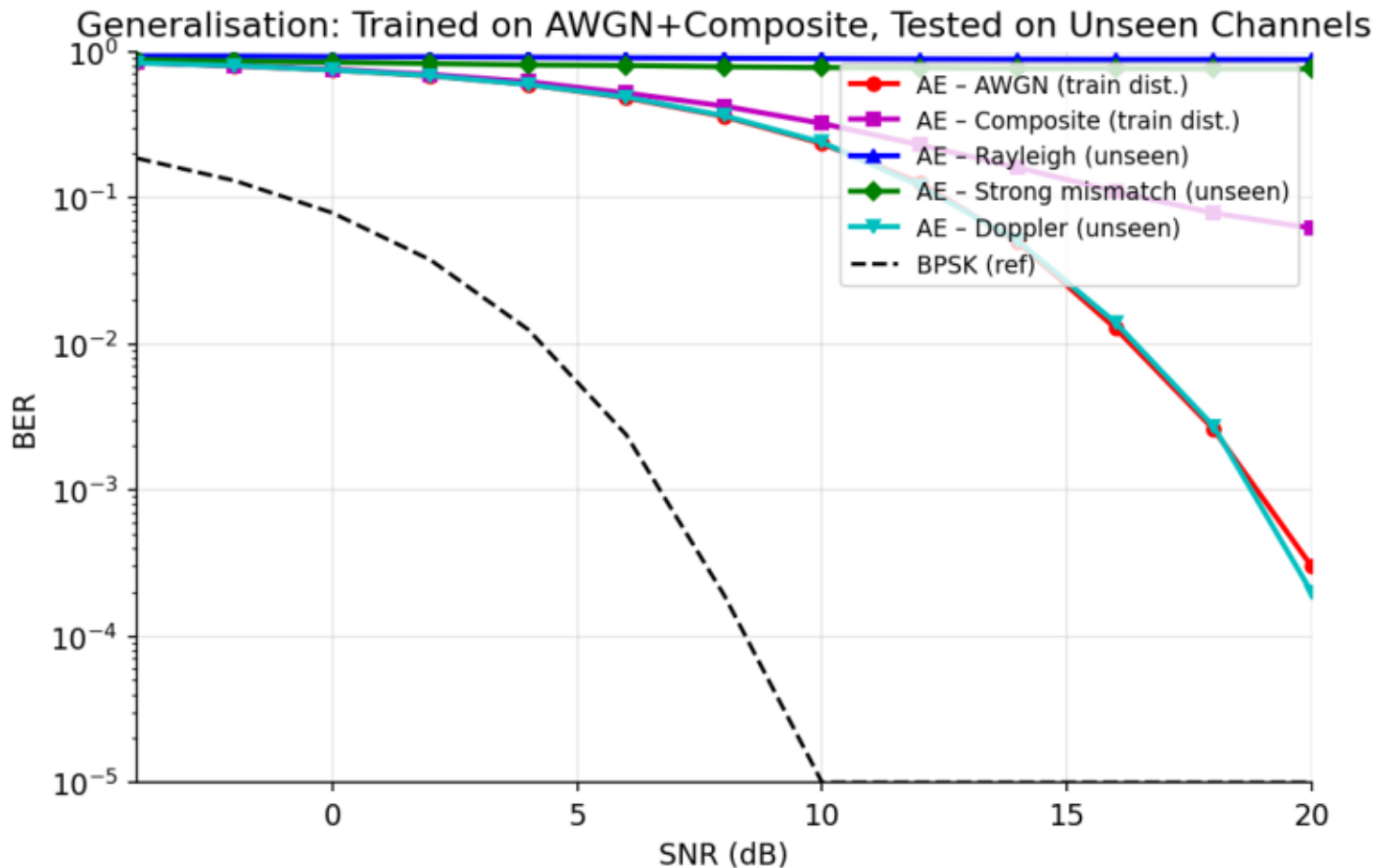


Fig 4. Non-Gaussian Noise Robustness

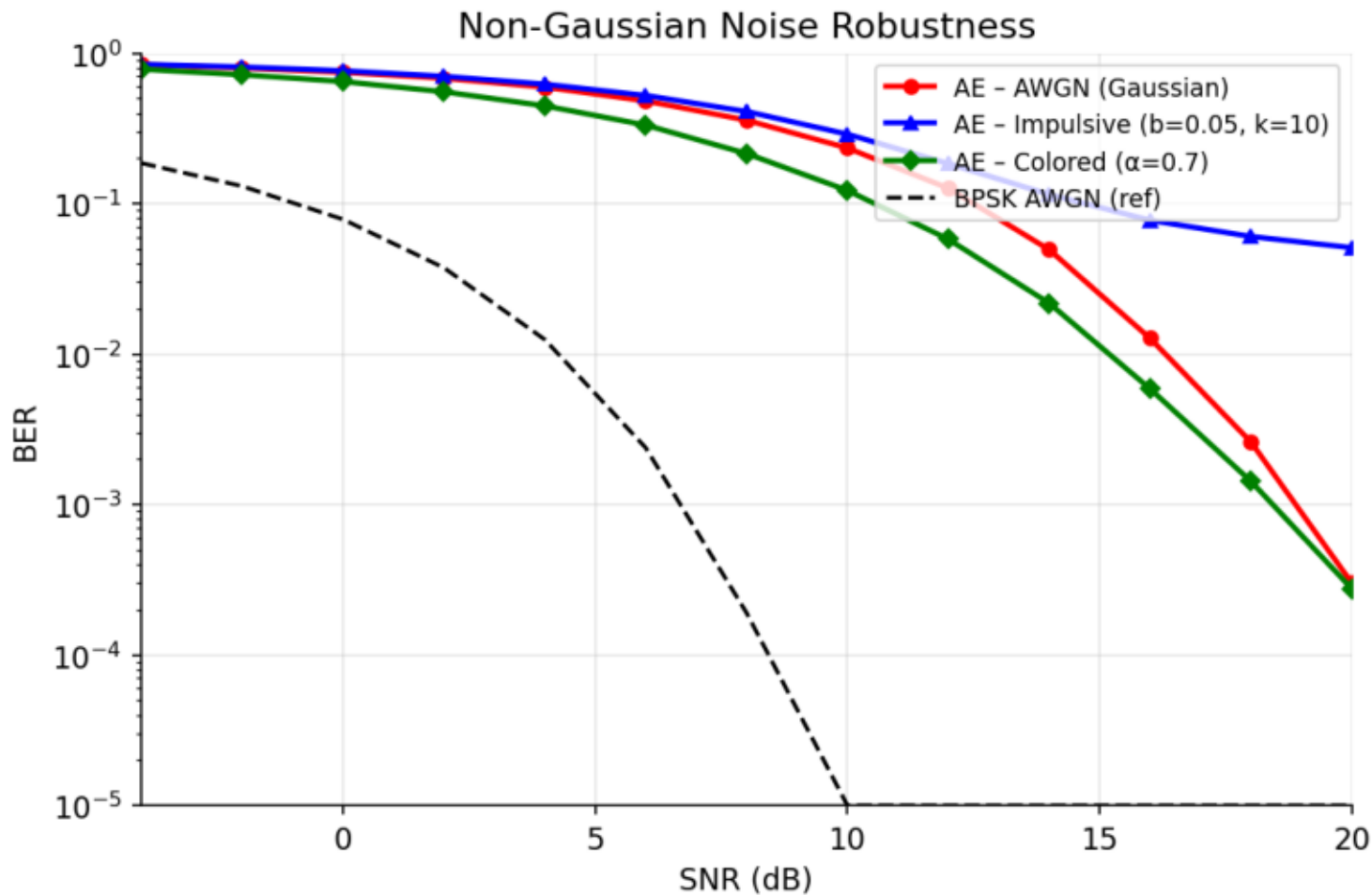


Fig 5. Learned Constellation (no noise)

Learned Constellation (M=16, clean)

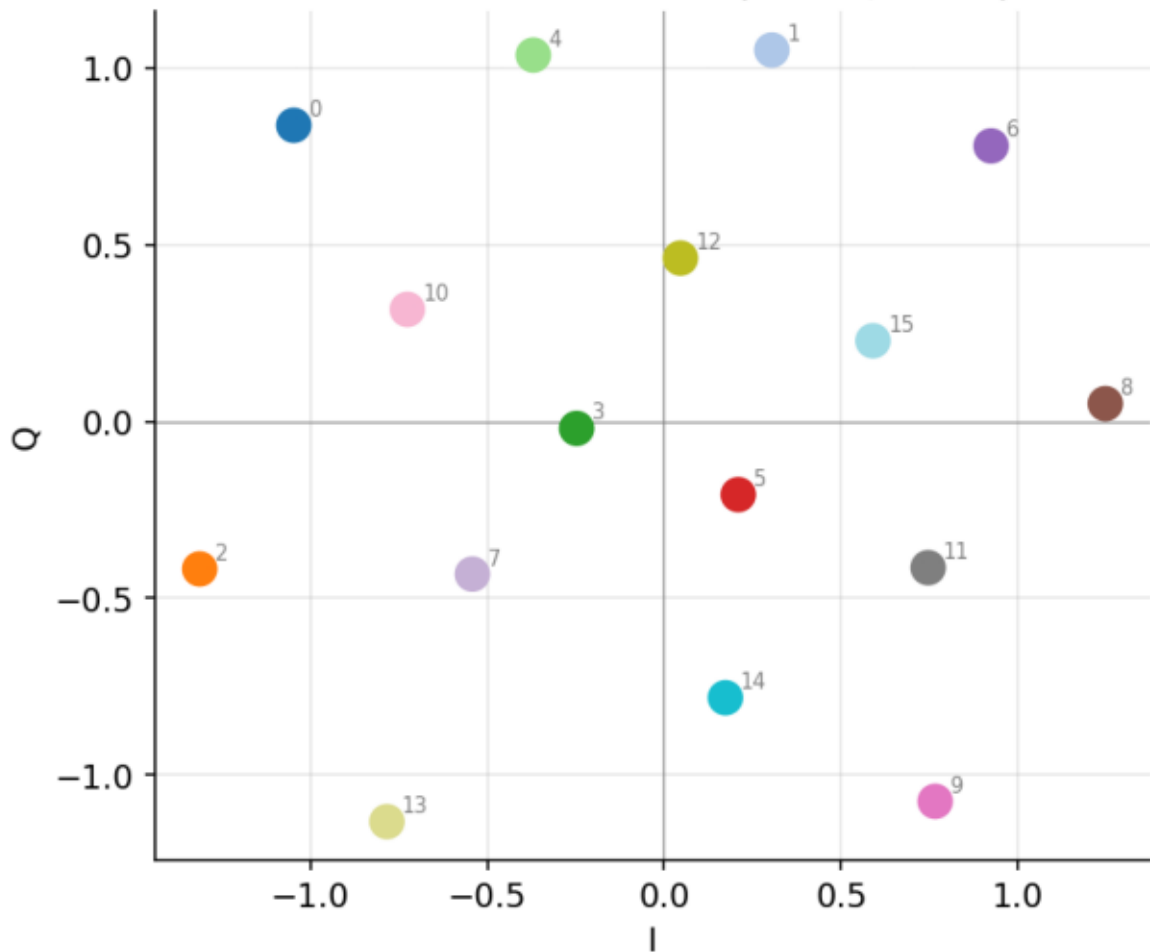


Fig 6. Learned Constellation under Composite Channel (SNR=10 dB)

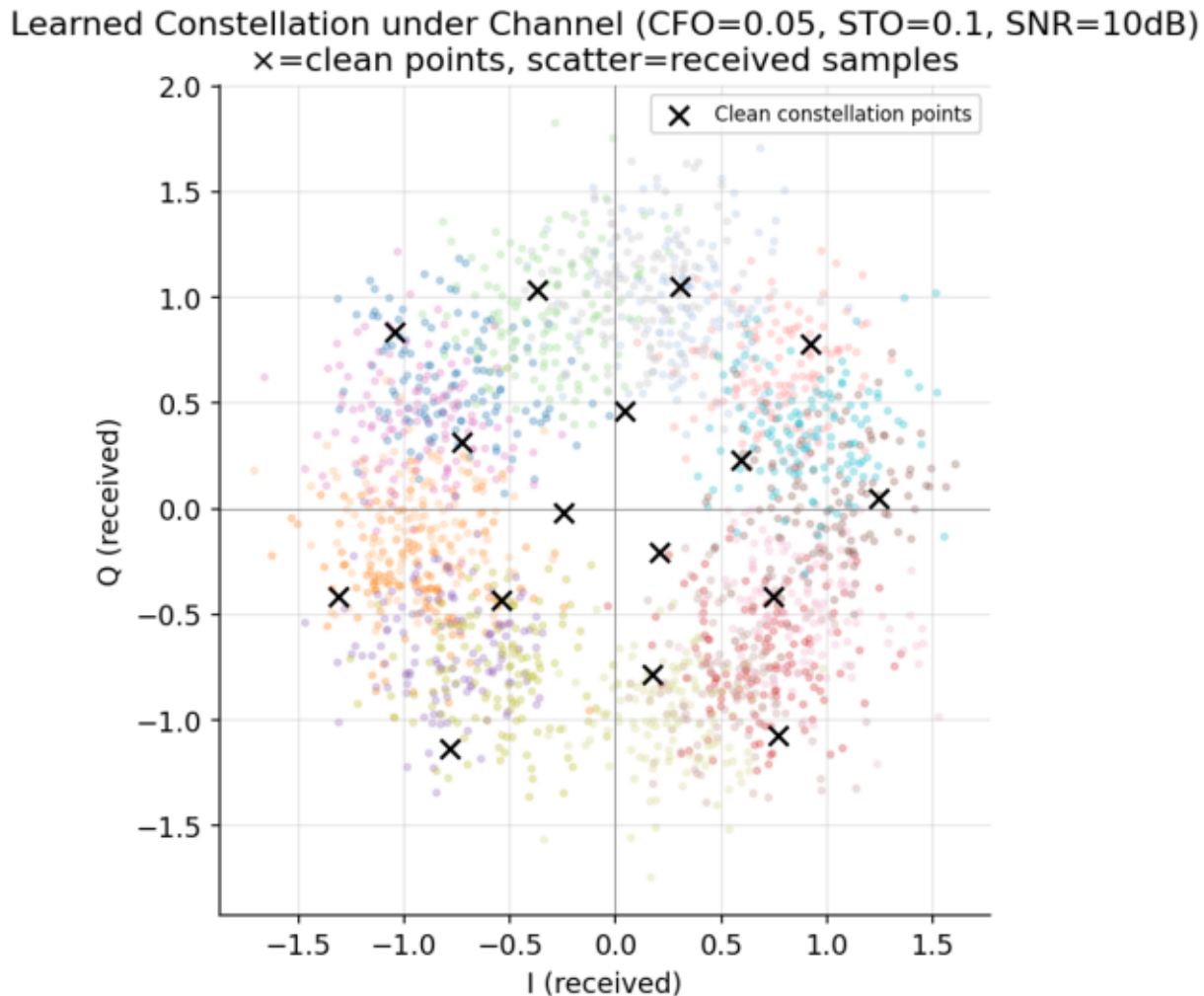


Fig 7. Training Loss and Accuracy Curves

Autoencoder Training Curves (3-Phase)

